(a-c) **Textual** Complexity Evolution

(d) **Temporal Visual** Complexity Evolution

Open-Vocabulary Segmentation:



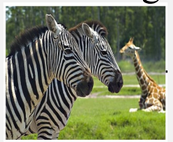
Tree
Buildings
...

Open-Vocabulary Segmentation LVLM



a

Referring Segmentation:



the second zebra from the front

Referring Segmentation LVLM



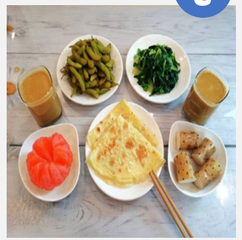
b

Reasoning Segmentation:



What is the food with the most Vitamin C in this image?

Reasoning Segmentation LVLM



c

d.1 VLM can Segment ***Single***-temporal! ***can't*** Segment Bi-temporal!







User: Could you Segment {*the parked car next to the sidewalk that disappeared canvas*}?

Single-temporal MLLM

Segmenter: <Mask>.

Results are generated by open-source checkpoint LISA-7B-v1

d.2 ***How*** to Segment Bi-temporal ?



Failure to distinguish Bi-temporal differences

Propose task **MTRS** for multi-temporal inputs

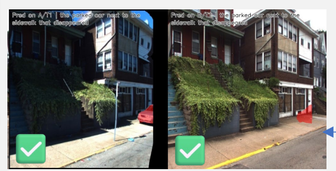
Label **MTRefSeg-21K** Dataset
Achieve Bi-temporal for VLM

Design Efficient **Change-aware** VLM for Bi-temporal Seg.

d.3 VLM can Segment Bi-temporal Input Using ***MTRefSeg-21K***







User: Could you Segment {*the parked car next to the sidewalk that disappeared canvas*}?

Multi-temporal MLLM

Segmenter: <Mask>.

Results are generated by fine-tuned checkpoint LISA-7B-v1-Bifusion